

- 1 A single sheet of aluminium foil is folded twice to produce a stack of four sheets. The total thickness of the stack of sheets is measured to be (0.80 ± 0.02) mm. This measurement is made using a digital calliper with zero error of (-0.20 ± 0.02) mm.

What is the percentage uncertainty in the calculated thickness of a single sheet?

- A** 1.0% **B** 2.0% **C** 4.0% **D** 6.7%

Ans: C

$$4t = (1.00 \pm 0.04) \text{ mm}$$

$$t = (0.25 \pm 0.01) \text{ mm}$$

$$\begin{aligned} \frac{\Delta t}{t} \times 100\% &= \frac{0.01}{0.25} \times 100\% \\ &= 4.0\% \end{aligned}$$

- 2 A voltmeter connected across a resistor in a circuit gives readings which have high precision